



HW ASSIGNMENT #3

LCA Exercise

Option B: Single-Use Cup vs. Reusable Steel Cup
at the Karnaf Store, Reichman University

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1. Functional Unit

The functional unit is: **serving one hot drink per day over 20 consecutive days at the Karnaf store, Reichman University** (20 servings total).

This unit allows direct comparison between the two systems: 20 single-use paper cups with polystyrene (PS) lids (Product A) versus one reusable stainless steel cup with silicone lid and silicone heat-resistant wrapper, used 20 times (Product B).

2. System Boundaries Type

Both products are assessed using a **cradle-to-grave** system boundary approach, with the following stage coverage per product:

	Product A - Single-Use Cup	Product B - Reusable Steel Cup
Raw material production	Included	Included
Transport to production site	Included	Included
Manufacturing	Included	Included
Transport to Reichman University	Excluded (not in assignment scope)	Included (explicitly stated)
Use phase (washing)	Not applicable	Included (20 washes)
End-of-life / disposal	Included (incineration after each use)	Excluded (per assignment scope)
Coffee production / store operations	Excluded	Excluded

The main structural asymmetry is that Product A includes disposal (cups are discarded after each use) while Product B excludes end-of-life per the assignment instructions. This is noted as a limitation when comparing the two totals on a fully equivalent basis.

3. System Boundary Diagrams

Product A - Single-Use Paper Cup (20 cups)

STAGE	ACTIVITY	IN / OUT OF SCOPE
Raw materials	Paper (Hedera) LDPE (Haifa) PS (Haifa)	IN
Transport	Lorry to production site in Natanya (~38-63 km)	IN
Manufacturing	Cup forming (0.04 kWh/cup, IL grid) LDPE coating application	IN
Transport to RU	Natanya to Herzliya (~30 km)	OUT (not in scope)
Use	20 cups used, 1 per day over 20 days	IN (defines scale)
End-of-life	Municipal incineration after each use (paper + LDPE + PS)	IN
Excluded	Coffee, filling, water, store operations, ink	OUT

Product B - Reusable Steel Cup (1 cup, 20 uses)

STAGE	ACTIVITY	IN / OUT OF SCOPE
Raw materials	Steel 18/8 (Espoo) Silicone lid (Espoo) Silicone wrapper (Espoo)	IN
Transport (FI)	Lorry: Espoo to Helsinki (~20 km)	IN
Manufacturing	Cold impact extrusion (steel) Silicone shaping (0.4 kWh/kg, FI grid)	IN
Transport to RU	Container ship: Helsinki to Herzliya (~7,500 km)	IN
Use phase	20 washes: 0.7 L water + 5 g soap per wash	IN
End-of-life	Not assessed (excluded per assignment)	OUT
Excluded	Coffee, store operations, transport store to user	OUT

4. Main Assumptions

Transport distances (estimated):	Hedera to Natanya: ~38 km; Haifa to Natanya: ~63 km; Espoo to Helsinki: ~20 km; Helsinki to Herzliya by sea: ~7,500 km (via Baltic Sea, North Sea, English Channel, Strait of Gibraltar, Mediterranean).
Lorry dataset - Israel proxy:	No Israel-specific lorry transport dataset exists in the BAFU database. The European average (RER) fleet average for 16t-32t lorries is used as a proxy. This is standard LCA practice when the exact country is unavailable.
Paper material:	Kraft paper, unbleached {RER} was selected as the closest available proxy for the paper cup body. No dedicated food-service paperboard or paper cup dataset exists in the BAFU 2025 database.
Polystyrene grade:	General Purpose Polystyrene (GPPS) {RER} was selected for the injection-molded PS lid, as GPPS is the standard grade used for rigid food packaging lids.
Steel grade:	Chromium steel 18/8 {RER} (i.e., 304 stainless steel) was selected for the reusable cup body. This is the industry-standard grade for reusable drinkware. The RER geography is used as a proxy for Finnish production; no FI-specific stainless steel dataset exists in BAFU.
Finland electricity:	The Electricity, production mix {FI} dataset (0.102 kgCO ₂ eq/kWh) was used, representing the national grid average. Finland's grid is predominantly nuclear and hydropower, which explains the low factor.
Tap water - Israel proxy:	The RER (European average) tap water dataset was used as no Israel-specific dataset exists. Israel relies heavily on desalinated water (~9.68x10 ⁻³ kgCO ₂ eq/kg) vs RER (2.03x10 ⁻⁴ kgCO ₂ eq/kg). Given water's small share of total GWP, this underestimation is negligible.
LDPE coating emission:	The process emission of 0.002 kgCO ₂ eq/cup was provided directly in the assignment and applied as a direct emission, not linked to a BAFU dataset.
Disposal (Product A):	All 20 cups and lids are sent to municipal incineration. Coated paper cups cannot enter standard paper recycling streams due to the plastic lining, making incineration the appropriate end-of-life route.
No processing losses (Product A):	The assignment explicitly states no waste is generated during cup production.
Cold impact extrusion (Product B):	The BAFU dataset 'Deformation stroke, cold impact extrusion, steel {RER}' (3.36x10 ⁻² kgCO ₂ eq/kg) was applied per kg of steel formed. This covers forming energy only; steel raw material is accounted for separately.
Transport Natanya to RU (Product A):	Excluded from scope, as the assignment defines Product A's transport only as materials to production site. The impact of including it would be ~0.002 kgCO ₂ eq (0.2% of total A), which is negligible.

5. Total GWP Results

Product A - Single-Use Cup (20 cups, 20 lids)

Life Cycle Stage	GWP [kgCO2eq]	Contribution [%]	Notes
Raw materials	0.510	42.6%	Paper + LDPE + Polystyrene (PS dominates)
Transport (to production site)	0.003	0.3%	Short intra-Israel distances
Processing	0.358	29.9%	Cup forming electricity + LDPE coating
Disposal (incineration)	0.327	27.3%	Paper + LDPE + PS after each use
TOTAL	1.197	100%	

Product B - Reusable Steel Cup (1 cup, 20 uses)

Life Cycle Stage	GWP [kgCO2eq]	Contribution [%]	Notes
Raw materials	0.775	79.6%	Steel dominates; silicone secondary
Transport	0.024	2.4%	Sea freight Helsinki to Herzliya
Processing (manufacturing)	0.007	0.7%	Low due to Finland's clean grid
Use phase (20 washes)	0.169	17.3%	Soap contributes 17.0%; water negligible
TOTAL	0.974	100%	End-of-life excluded per assignment



The reusable cup has approximately **19% lower GWP** over the 20-day comparison period. Importantly, Product B's manufacturing emissions are fixed regardless of how many times the cup is used. With additional uses, the gap widens further in favor of the reusable option.

6. Largest GWP Contributor

Product A - Polystyrene (PS)

The largest single contributor is **polystyrene (PS)**. Its raw material production contributes 0.292 kgCO2eq (24.4% of total), and its incineration at end-of-life adds another 0.256 kgCO2eq (21.4%). Combined, the PS lid accounts for **45.8%** of Product A's total GWP - nearly half the entire footprint of 20 disposable cups. Cup-forming electricity is the next largest single item at 26.5%, reflecting Israel's relatively carbon-intensive grid (0.397 kgCO2eq/kWh).

Flow	GWP [kgCO ₂ eq]	% of Total A
PS raw material production	0.292	24.4%
Cup forming electricity	0.318	26.5%
PS incineration (disposal)	0.256	21.4%
Paper raw material	0.171	14.3%
LDPE disposal	0.061	5.1%
LDPE coating process	0.040	3.3%
LDPE raw material	0.047	3.9%
Transport	0.003	0.3%
Paper disposal	0.010	0.9%

Product B - Stainless Steel (18/8)

Steel production dominates at **68.1%** of total GWP (0.663 out of 0.974 kgCO₂eq). This reflects the energy-intensive nature of stainless steel manufacturing. Soap consumption over 20 washes is the second-largest contributor at 17.0% - notable given that only 100 g of soap is used in total. This underscores that the use-phase cleaning routine, rather than manufacturing, drives the ongoing emissions of the reusable product.

Flow	GWP [kgCO ₂ eq]	% of Total B
Steel production (18/8)	0.663	68.1%
Soap (20 washes)	0.166	17.0%
Silicone lid production	0.070	7.2%
Silicone wrapper production	0.042	4.3%
Sea transport (Helsinki to Herzliya)	0.023	2.3%
Water (20 washes)	0.003	0.3%
Cold impact extrusion	0.005	0.5%
Processing (FI electricity)	0.002	0.2%
Land transport (Espoo to Helsinki)	0.001	0.1%

Appendix - Inventory Table: Product A (Single-Use Cup)

Material / flow / activity	Qty	Unit	BAFU Dataset	Geo	EF [kgCO ₂ eq/unit]	GWP [kgCO ₂ eq]	%	Notes / assumptions
Paper (cup body)	0.240	kg	Kraft paper, unbleached, at plant	RER	7.11E-01	0.1707	14.3%	20 cups x 0.012 kg. No paper-cup dataset in BAUFU; unbleached kraft used as proxy.
LDPE (coating)	0.020	kg	Polyethylene, LDPE, granulate, at plant	RER	2.35E+00	0.0470	3.9%	20 cups x 0.001 kg.
Polystyrene (lid)	0.080	kg	Polystyrene, general purpose, GPPS, at plant	RER	3.65E+00	0.2920	24.4%	20 lids x 0.004 kg. GPPS selected for rigid injection-molded lid.
Transport - Paper (Hedera to Natanya)	0.00912	tkm	Transport, freight, lorry, 16t-32t gross weight, fleet average	RER	2.22E-01	0.0020	0.2%	0.240 kg x 38 km. RER proxy for Israel; no IL-specific dataset.
Transport - LDPE (Haifa to Natanya)	0.00126	tkm	Transport, freight, lorry, 16t-32t gross weight, fleet average	RER	2.22E-01	0.0003	0.0%	0.020 kg x 63 km.
Transport - PS (Haifa to Natanya)	0.00504	tkm	Transport, freight, lorry, 16t-32t gross weight, fleet average	RER	2.22E-01	0.0011	0.1%	0.080 kg x 63 km.
Cup forming (Israel electricity)	0.800	kWh	Given in assignment (IL grid)	IL	3.97E-01	0.3176	26.5%	20 cups x 0.04 kWh. Israel grid emission factor given directly.
LDPE coating process	20	cups	Given in assignment (direct emission)	IL	2.00E-03/cup	0.0400	3.3%	Direct process emission as specified in assignment.
Disposal - Paper (incineration)	0.240	kg	Disposal, packaging paper, 13.7% water, to municipal incineration	CH	4.33E-02	0.0104	0.9%	Coated paper cannot be recycled; incineration assumed. CH proxy for IL.
Disposal - LDPE (incineration)	0.020	kg	Disposal, polyethylene, 0.4% water, to municipal incineration	CH	3.03E+00	0.0606	5.1%	LDPE coating disposed with cup body. CH proxy for IL.
Disposal - PS (incineration)	0.080	kg	Disposal, polystyrene, 0.2% water, to municipal incineration	CH	3.20E+00	0.2560	21.4%	20 PS lids after use. CH proxy for IL.
TOTAL						1.197	100%	

Appendix - Inventory Table: Product B (Reusable Steel Cup)

Material / flow / activity	Qty	Unit	BAFU Dataset	Geo	EF [kgCO ₂ eq/unit]	GWP [kgCO ₂ eq]	%	Notes / assumptions
Steel (cup body)	0.150	kg	Chromium steel 18/8, at plant	RER	4.42E+00	0.6630	68.1%	304 stainless steel (18/8). RER used; no FI-specific stainless steel dataset in BAUFU.
Silicone (lid)	0.025	kg	Silicone product, at plant	RER	2.79E+00	0.0698	7.2%	
Silicone (wrapper)	0.015	kg	Silicone product, at plant	RER	2.79E+00	0.0419	4.3%	
Transport - land (Espoo to Helsinki)	0.00380	tkm	Transport, freight, lorry, 16t-32t gross weight, fleet average	RER	2.22E-01	0.0008	0.1%	0.190 kg total x 20 km.
Transport - sea (Helsinki to Herzliya)	1.4250	tkm	Transport, transoceanic container ship	OCE	1.60E-02	0.0228	2.3%	0.190 kg x 7,500 km. Route via North Sea, English Channel, Mediterranean.

Material / flow / activity	Qty	Unit	BAFU Dataset	Geo	EF [kgCO ₂ eq/unit]	GWP [kgCO ₂ eq]	%	Notes / assumptions
Silicone lid shaping (Finland electricity)	0.0100	kWh	Electricity, production mix	FI	1.02E-01	0.0010	0.1%	0.4 kWh/kg x 0.025 kg.
Silicone wrapper shaping (Finland electricity)	0.0060	kWh	Electricity, production mix	FI	1.02E-01	0.0006	0.1%	0.4 kWh/kg x 0.015 kg.
Cold impact extrusion (steel cup)	0.150	kg	Deformation stroke, cold impact extrusion, steel	RER	3.36E-02	0.0050	0.5%	Forming process only; steel material accounted separately above.
Tap water (20 washes)	14.00	kg	Tap water, at user	RER	2.03E-04	0.0028	0.3%	14 L total (0.7 L x 20). RER proxy; IL uses desalination (slightly higher factor, negligible impact).
Soap (20 washes)	0.100	kg	Soap, at plant	RER	1.66E+00	0.1660	17.0%	5 g/wash x 20 washes = 100 g.
TOTAL						0.974	100%	End-of-life excluded per assignment scope.

AI Disclosure: Claude (Anthropic, claude-sonnet-4-6) was used as a brainstorming partner and calculation aid: helping identify relevant BAFU datasets, verifying numerical calculations, and suggesting how to structure the results. All analytical decisions - including dataset selection, geographic assumptions, system boundary choices, and interpretation of results - were made by Omri Shamgar, Tom Ginel, and Tomer Tasa. The written conclusions reflect the students' own understanding of the material.